KISSsoft 2026 native report - readable text rendering
Source: R1prime-gear80-coupling1.pprpt
No recalculation. Graphic payloads omitted; retain native PPRPT for full layout.

Shaft
R1prime-gear80-coupling1
Changed on | 31.10.2026, 16:53:01 |
Changed by | AAA |
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Description | P4 candidate - STATIC PREP ONLY - loads/supports must be verified before calculation |
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 | 2o2dy (5416) |
 | KISSsoft Release 2026 |
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1 Messages
| Calculation is consistent. |
2 Input data
Coordinate system shaft: see picture W-002
2.1 Shafts
2.1.1 Shaft 'Shaft'
Drawing W-007
Initial position (mm) -10.000
Length (mm) 170.000
Speed (1/min) 900.000
Direction of rotation: clockwise
Material, own Input P4 custom DIN743 1000/800/500/400/300
Young's modulus (N/mm²) 206000.000
Poisson's ratio nu 0.300
Density (kg/m³) 7830.000
Coefficient of thermal expansion (10^-6/K) 11.500
Temperature (°C) 20.000
Mass of shaft (kg) 0.982
Mass of shaft, including additional masses (kg) 0.982
Mass moment of inertia (kg*mm²) 122.100
Momentum of mass GD2 (Nm²) 0.005
2.2 Weight force
The direction of the weight is not considered
2.3 Shaft modeling
2.4 Shear deformations
Consider deformations due to shearing
Shear correction factor 1.100
2.5 Rolling bearings
Rolling bearing stiffness is calculated from inner bearing geometry
Tolerance field: Mean value
2.6 Housing
Reference temperature (°C) 20.000
2.7 Load applications
3 Shaft definition (Shaft)
3.1 Outer contour
3.1.1 Cylinder (Cylinder 1), 0.000mm ...20.000mm
Diameter (mm) [d] 30.0000

Length (mm) [l] 20.0000
Surface roughness (µm) [Rz] 4.8000
3.1.2 Cylinder (Cylinder 2), 20.000mm ...100.000mm
Diameter (mm) [d] 34.0000
Length (mm) [l] 80.0000
Surface roughness (µm) [Rz] 16.0000
3.1.3 Cylinder (Cylinder 3), 100.000mm ...120.000mm
Diameter (mm) [d] 30.0000
Length (mm) [l] 20.0000
Surface roughness (µm) [Rz] 4.8000
3.1.4 Cylinder (Cylinder 4), 120.000mm ...170.000mm
Diameter (mm) [d] 25.0000
Length (mm) [l] 50.0000
Surface roughness (µm) [Rz] 4.8000
3.2 Forces
3.2.1 Coupling (Coupling / Motor)
Position on shaft (mm) [ylocal] 150.0000
Position in global system (mm) [yglobal] 140.0000
Effective diameter (mm) [d] 0.0000
Radial force factor (-) [K2] 0.0000
Direction of the radial force (°) [ρ] 0.0000
Axial force factor (-) [K3] 0.0000
Length of load application (mm) [l] 1.0000
Power (kW) [P] 2.3562 driving (output)
Torque (Nm) [T] -25.0000
Axial force (N) [Fy] 0.0000
Radial force X (N) [Fx] 0.0000
Radial force Z (N) [Fz] 0.0000
Bending moment X (Nm) [Mx] 0.0000
Bending moment Z (Nm) [Mz] 0.0000
Mass (kg) [M] 0.0000
Mass moment of inertia (kg*m²) [Jp] 0.0000e+00
Mass moment of inertia around X-axis (kg*m²) [Jxx] 0.0000e+00
Mass moment of inertia around Z-axis (kg*m²) [Jzz] 0.0000e+00
3.2.2 Cylindrical gear (Cylindrical gear)
Position on shaft (mm) [ylocal] 60.0000
Position in global system (mm) [yglobal] 50.0000
Operating pitch diameter (mm) [dw] 40.0000
Spur gear
Gear geometry: symmetric
Working pressure angle at normal section (°) [αwn] 20.0000
Position of contact (°) [αpos] 180.0000
Length of load application (mm) [l] 80.0000
Power (kW) [P] 2.3562 driven (input)
Torque (Nm) [T] 25.0000
Axial force (N) [Fy] 0.0000
Radial force X (N) [Fx] 454.9628
Radial force Z (N) [Fz] 1250.0000
Bending moment X (Nm) [Mx] -0.0000
Bending moment Z (Nm) [Mz] -0.0000
3.3 Bearing
3.3.1 P4 Support A (Own input)
Bearing position (mm) [ylokal] 10.000
Bearing position (mm) [yglobal] 0.000
Degrees of freedom
X: fixed; Y: fixed; Z: fixed;
Rx: free; Ry: free; Rz: free;
3.3.2 P4 Support B (Own input)
Bearing position (mm) [ylokal] 110.000
Bearing position (mm) [yglobal] 100.000
Degrees of freedom
X: fixed; Y: free; Z: fixed;
Rx: free; Ry: free; Rz: free;
4 Gears
Gears are considered only as load application.
5 Results
5.1 Shafts
Maximum deflection (µm) 2.999
Position of the maximum (mm) 160.000
Mass center of gravity (mm) 76.630
Total axial load (N) 0.000
Torsion of the shaft under torque (°) -0.024
(Difference between left and right shaft end)
Nominal stresses, without taking into account stress concentrations
GEH(von Mises): sigV = ((sigB+sigZ,D)^2 + 3*(tauT+tauS)^2)^1/2
SSH(Tresca): sigV = ((sigB-sigZ,D)^2 + 4*(tauT+tauS)^2)^1/2
5.2 Bearing
5.3 'Shaft' Bearing 'P4 Support A'

Position (Y-coordinate) [y] 10.00 mm
Bearing reaction force [Fx] -0.227 kN
Bearing reaction force [Fy] -0.000 kN
Bearing reaction force [Fz] -0.625 kN
Bearing reaction force [Fr] 0.665 kN
Inclination angle [αFr] -110.000 °
Displacement of bearing [ux] 0.000 µm
Displacement of bearing [uy] 0.000 µm
Displacement of bearing [uz] 0.000 µm
Displacement of bearing [ur] 0.000 µm
Misalignment of bearing [rx] 0.047 mrad
Misalignment of bearing [rz] -0.017 mrad
Misalignment of bearing [rr] 0.050 mrad
5.4 'Shaft' Bearing 'P4 Support B'
Position (Y-coordinate) [y] 110.00 mm
Bearing reaction force [Fx] -0.227 kN
Bearing reaction force [Fy] 0.000 kN
Bearing reaction force [Fz] -0.625 kN
Bearing reaction force [Fr] 0.665 kN
Inclination angle [αFr] -110.000 °
Displacement of bearing [ux] 0.000 µm
Displacement of bearing [uy] -0.000 µm
Displacement of bearing [uz] 0.000 µm
Displacement of bearing [ur] 0.000 µm
Misalignment of bearing [rx] -0.047 mrad
Misalignment of bearing [rz] 0.017 mrad
Misalignment of bearing [rr] 0.050 mrad
6 Strength calculation according to DIN 743:2012
6.1 Summary
Shaft
Drawing W-007
Material P4 custom DIN743 1000/800/500/400/300
Material type Through hardened steel
Material treatment alloyed/through hardened
Surface treatment No
Calculation of endurance limit and the static strength
Calculation for load case 1 (σmv = const)
Cross section Position (Y-Coord) (mm)
S2 65.00 Smooth shaft
Results:

Cross section	βσ	KFσ	K2d	SD	SS
S2	1.00	0.88	0.90	71.76	70.97

Required safeties: 1.20 1.20
Abbreviations:
βσ: Notch factor, bending
KFσ: Surface factor
K2d: size factor bending
SD: Safety endurance limit
SS: Safety against yield point
6.1.1 Utilization (%)

Cross section	Static (yield point)	Endurance
S2	1.691	1.672

Maximum utilization (%) [A] 1.691
Utilization = Smin/S (%)
6.2 Calculation details
General statements
Label Shaft
Drawing W-007
Length (mm) [l] 170.00
Speed (1/min) [n] 900.00
Material P4 custom DIN743 1000/800/500/400/300
Material type Through hardened steel
Material treatment alloyed/through hardened
Surface treatment No

Tension/Compression	Bending	Torsion	Shearing
Load factor static calculation	2.000	2.000	2.000	2.000
Stress ratio	0.000	-1.000	1.000	-1.000
Load factor endurance limit	1.000	1.000	1.000	1.000

Reference diameter material (mm) [dB] 16.00
σB according to DIN 743 (at dB) (N/mm²) [σB] 1000.00
σB according to DIN 743 (at dB) (N/mm²) [σS] 800.00
[σzdW], bei dB (N/mm²) 400.00
[σbW], bei dB (N/mm²) 500.00
[τtW], bei dB (N/mm²) 300.00
Thickness of raw material (mm) [dWerkst] 34.00
Material data (σB, σS, σW) according to DIN 743/3 for reference diameter
Material strength values for shaft diameter calculated with size factor K1(deff)
Material strength calculated from size of raw material

Geometric size factor K1(deff) calculated from raw material diameter
[σBeff] (N/mm²) 914.89
[σSeff] (N/mm²) 710.96
[σbFK] (N/mm²) 853.15
[τtFK] (N/mm²) 492.57
[σbBRand] (N/mm²) 708.00
[σzdW] (N/mm²) 365.95
[σbW] (N/mm²) 457.44
[τtW] (N/mm²) 274.47
Endurance limit for single stage use
Calculation for load case 1 (σ.mv = const)
6.2.1 Cross section 'S2' Smooth shaft
Comment M36 R1 independent section, local coordinate
Position (Y-Coordinate) (mm) [y] 65.000
External diameter (mm) [da] 34.000
Inner diameter (mm) [di] 0.000
Notch effect Smooth shaft
Mean roughness (µm) [Rz] 6.300
Tension/Compression Bending Torsion Shearing
Load: (N) (Nm)
Mean value [Fzdm, Mbm, Tm, Fqm] 0.0 0.0 14.1 0.0
Amplitude [Fzda, Mba, Ta, Fqa] 0.0 19.7 0.0 83.3
Maximum value [Fzdmax, Mbmax, Tmax, Fqmax] 0.0 19.7 14.1 83.3
Maximum value (static safety) 0.0 39.5 28.1 166.6
Cross section, moment of resistance: (mm²)
[A, Wb, Wt] 907.9 3858.7 7717.3
Stresses: (N/mm²)
[σzdm, σbm, τm, τqm] (N/mm²) 0.000 0.000 1.822 0.000
[σzda, σba, τa, τqa] (N/mm²) 0.000 5.115 0.000 0.122
[σzdmax,σbmax,τmax,τqmax] (N/mm²) 0.000 10.230 3.644 0.245
Technological size influence [K1(σB)] 0.915
[K1(σS)] 0.889
Tension/Compression Bending Torsion
Notch effect coefficient [ß] 1.000 1.000 1.000
Geometrical size influence [K2(d)] 1.000 0.899 0.899
Influence coefficient surface roughness [KF] 0.884 0.884 0.933
Own input of
Surface treatment factor [KV] 1.000 1.000 1.000
Total influence coefficient [K] 1.131 1.244 1.184
Present safety for endurance limit:
Equivalent mean stress (N/mm²) [σmV] 3.156
Equivalent mean stress (N/mm²) [τmV] 1.822
Fatigue limit of part (N/mm²) [σWK] 323.459 367.834 231.855
Influence coefficient of mean stress sensitivity.
[ψσK] 0.215 0.252 0.145
Permissible amplitude (N/mm²) [σADK] 322.781 367.040 231.591
Safety against fatigue [S] 71.756
Required safety against fatigue [Smin] 1.200
Result (%) [S/Smin] 5979.7
Present safety
for proof against exceed of yield point:
Static notch sensitivity factor [K2F] 1.000 1.200 1.200
Increase coefficient [γF] 1.000 1.000 1.000
Yield stress of part (N/mm²) [σFK] 710.958 853.150 492.566
Safety yield stress [S] 70.972
Required safety [Smin] 1.200
Result (%) [S/Smin] 5914.3
Remarks:
- The shearing force is not considered in the analysis specified in DIN 743.
- Cross section with interference fit: The notch factors for the 'Slight interference fit' case are no longer defined in DIN 743.
  The notch factors are taken from the FKM Guideline, except if those for the 'tight interference fit' according to DIN are smaller,
in which case they will be used.
End of report (lines: 379)